

A Gaussian Mixture Hidden Markov Model for the VIX

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Submitted to *Quantitative Finance*, 22 June 2022

Revision, 3 December 2022

A Hidden Markov Model (HMM) is used to model the VIX (the Cboe Volatility Index). A 4-state Gaussian mixture is fitted to the VIX price history from 1990 to 2022. Using a growing window of training data, the price of the S&P500 is predicted and two trading algorithms are presented, based on the expected mean and standard deviation as well as the predicted state probabilities. The performance of these strategies is analyzed on the full data-set and compared to the results of a data-set of half the size. The strategies are compared to both a buy-n-hold strategy of the same period and the VXTH (the Cboe VIX Tail Hedge Index). The results presented here show promising application in modelling and predicting volatility, as well as identifying current volatility regimes predominating the market.

Keywords: Gaussian Mixture, Hidden Markov Model, VIX, Volatility, S&P500, Trading, Regimes, Bull, Bear, Market, Implied Volatility Surfaces

JEL: A10, C00, C01, C02, C11, C10, C15, C19, C22, C32, C38, C53, C58, C63, G17

1 – Introduction

The basic theory of Hidden Markov Models was developed in the 1960s and applied to speech processing applications in the early 1970s [1, 2]. It derives from the concept of a Markov chain developed by the Russian mathematician Andrey Markov (1856, †1922) in 1918, which forms the basis of what later became known as a stochastic process. A Markov chain in its simplest form is a stochastic process where the current state of the process only depends on one or more previous states and nothing else. An order one Markov process depends only on the previous state and is most commonly used. A Hidden Markov Model is a Markov process that is governed by another stochastic process. This governing stochastic process is unobservable, thus hidden from the observer, only the observable process is measurable. A very basic example to demonstrate this is an operator tossing one or more coins in a secret room [1]. The operator will not tell you which coin he is tossing and only disclose the result of the coin toss. The results of the sequence of coin tosses represent the observation sequence. If there is only one coin in this experiment then there is no hidden stochastic process. A Hidden Markov model is needed to explain when there are different biased coins that the operator picks from using another random process, the hidden stochastic process. In simple terms a Hidden Markov Model can be thought of as a doubly stochastic process.

Solving a Hidden Markov Model problem always consists of three steps: Encoding, Decoding and Learning (also known as Training). For each of them different algorithms were developed in the 1960s. The Encoding step is solved using the Forward-Backward procedure, the decoding step is solved using the Viterbi algorithm and the Learning (Training) step is solved using the Expectation Maximization algorithm.

Even though Markov was initially interested in applying Markov chains to the Russian language, the application to speech recognition only accelerated from the 1960s onwards [3]. Many papers and several books [4, 5] have been written with applications to speech, speaker identification and noise processing. Hidden Markov Models were subsequently applied to a multitude of areas where patterns within data exist spanning disciplines such as text- and sound processing, medicine, biology, bioinformatics, environmental studies, meteorology, oceanography and industrial applications. Other current areas of interest include big data, sentiment analysis, customer behavior analytics, intention estimation and network security.

The other extensive topic that Hidden Markov Models are applied to, which is the primary interest of this paper is the stock market. Since it was known that stock markets undergo certain business cycles [6-8] and exhibit regimes [9] the application of a pattern recognition algorithm such as the Hidden Model in the 1990s was an obvious candidate to use [10-13]. The regimes of the stock market can be thought of as a set of various biased coins in the coin-toss example above. One coin being biased towards negative outcomes and other coins being biased to less negative and more positive outcomes. Simply put the Hidden Markov Model will describe which of the coins are in play currently. The stock market can be regarded as the hidden stochastic process that makes the movements of it seem random, which is a very convenient aspect of the Hidden Markov Model, since it introduces stochasticity automatically. It is worth noting here that Hidden Markov models were originally considered for discrete problems (such as the coin-toss), but they were later generalized to continuous problems.

There exist several books on Hidden Markov models in Finance [14-16] as well as a number of theses written [17-26]. Hidden Markov Models in finance are used to identify regimes [23, 27-39], predicting price moves [21, 40-52] and to model stochastic volatility itself [53-55]. Other applications can be found in mortgage delinquencies and credit behavior [56-62].

The topic of modelling volatility is a very extensive field of research, see [63]. The most researched models for volatility are auto-regressive moving average models (ARMA), autoregressive conditional heteroscedasticity-type models (ARCH, GARCH, etc.) and stochastic volatility models. In practice the ARCH-type models are most commonly used, since they allow for volatility clustering which is commonly observed in the data of volatility. All these approaches take the asset itself to be the source for the data and fit the models to it using some least squares method or similar. Further advances in research have been made into applying similar auto-regressive models to volatility assets themselves, such as the VIX (the Volatility Index) of the Chicago Board Options Exchange (Cboe), see [64, 65] and the references cited therein.

The VIX is the most commonly traded volatility asset globally and trades several hundred thousand contracts in terms of options, futures and several millions as Exchange Traded Funds and Notes daily, but there are also other volatility indices such as the Nasdaq-100 (VXN), Dow Jones Industrial Average (VXD) and Russell 2000.

In this paper we will use the Hidden Markov Model fitted to the VIX, providing another method to model volatility assets directly. Since the price of the VIX is continuous the Hidden Markov Model is generalized to the continuous case, which is done through replacing the discrete probability distribution with a mixture of continuous probability distribution functions. The most commonly used one is a Gaussian mixture, a series of normal distributions, since it provides the ability to model non-normal stochastic assets exhibiting skew, kurtosis or other forms of non-normality, while still providing the convenience of normality. Based off this model two trading strategies on the S&P500 are presented.

This paper is structured as follows. Section 2 describes the VIX and its relationship to the S&P500. Section 3 introduces the Hidden Markov Model and Gaussian mixture describing the main properties and a practical example. Section 4 covers the trading strategy employed utilizing the Gaussian Mixture Hidden Markov Model, followed by Section 5 the conclusions.

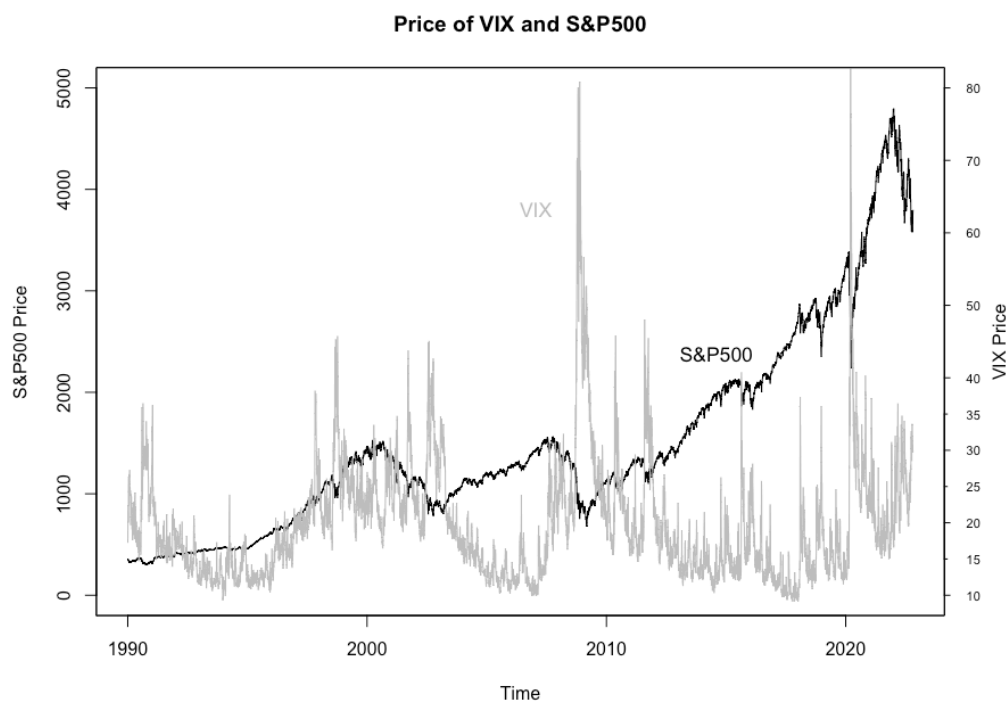


Figure 1: Chart of the S&P500 (lhs) and the VIX (rhs) from 1990 to 2022.

2 – VIX & S&P500

The VIX Index was first introduced in 1993 by Cboe Global Markets [66] and was based on the S&P100 Index initially, which was changed to the S&P500 Index in 2003. Since 2014 the calculation of the VIX includes weekly options too, since the rapid growth of the options market, weeklies making up a third of all S&P500 options traded.

The VIX is calculated as a time-weighted average of volatility between a pair of near-term T_1 and next-term T_2 volatilities σ_1 and σ_2

$$VIX = 100 \times \sqrt{\left\{ T_1 \sigma_1^2 \left[\frac{N_{T_2} - N_{30}}{N_{T_2} - N_{T_1}} \right] + T_2 \sigma_2^2 \left[\frac{N_{30} - N_{T_1}}{N_{T_2} - N_{T_1}} \right] \right\} \times \frac{N_{365}}{N_{30}}} \quad (1)$$

where N represents the number of minutes to settlement of the near-term N_{T_1} , next-term N_{T_2} options and number of minutes in a month N_{30} and year N_{365} .

Near-term corresponds to volatilities derived from option prices with 23 days and next-term with 37 days to expiry respectively. The average results in an interpolation of the 30-day volatility.

The volatilities σ_1 and σ_2 are calculated as an average over out-of-the-money (OTM) put and call options with an at-the-money strike price K_0 , using only options with a non-zero bid price. The range of OTM options may vary due to the market.

The near and next-term volatilities $\sigma_{1/2}$ are given by

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{R \cdot T} Q(K_i) - \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2 \quad (2)$$

with corresponding near and next-term values for T, R and F , where T represents the minutes to expiry scaled by the number of minutes in a year, R representing the risk-free interest rate to expiry and F the forward index prices derived from the options.

The forward index levels $F_{1/2}$ for the S&P500 are calculated using the put-call parity, with the strike of those options having the smallest price difference between calls and puts

$$F = \text{Strike} + e^{R \cdot T} \times (\text{Call Price} - \text{Put Price}) \quad (3)$$

$K_{0,1/2}$ are the strikes immediately below or equal to the forward index level $F_{1/2}$. Next, the contributions of all strikes lower than $K_{0,1/2}$ for put options are calculated, as long as the bids are non-zero. Similarly, for all strikes higher for all call options as long as there are non-zero bids. Once two or more bids are zero it stops. For $K_{0,1/2}$ the average of the put and calls are used, for all other strikes the midpoint price is used for either puts or calls. In Equation (2) the weighted sum of option contributions is calculated using the midpoint option prices $Q(K_i)$ weighted by the strike differences ΔK_i and the reciprocal of the strike prices squared $\frac{1}{K_i^2}$.

Formula (2) derives from a piece-wise linear approximation of a log contract replicated by a continuous strip of options representing variance. Demeterfi et.al [67] provide a derivation of this approximation. The VIX can therefore be regarded as the risk-neutral or fair value expectation of the 30-day realized volatility of the S&P500, calculated by an average of two volatilities, derived from theoretical values from a chain of market quotes for calls and puts at two expiries, which represent two variance swaps. These chains of calls and puts include all options quoted on the market with a non-zero bid, capturing the entire liquidity of the S&P500 options market. The chain only terminates when more than two consecutive strikes have zero bids.

The VIX has become very popular since its inception, since 2016 it has extended trading hours (3:00am to 9:15am ET) beyond the US market hours (9:30am to 4:15pm ET) and due to the nature of the volatility markets Cboe has introduced a filtering algorithm on the quoted spot VIX index levels.

The filter allows changes in spot index which are less than 0.49 volatility points, but excludes drops in spot index by more than 0.5 volatility points, until such move has persisted for more than 2 minutes. It is not affecting upward moves in spot index and only affects downward moves in volatility, making sure any bigger downward moves in volatility are persistent. Given the update to spot is allowed for downward moves of up to 0.49 volatility points this filter is only preventing wild downward drops in volatility and has no impact on spot moves upward and a negligible impact on spot moves in the VIX index overall.

Futures on the VIX were introduced in 2004, followed by options on the VIX in 2006, which became the most successful product launched by the Cboe. In 2021 a total options volume of 12.1 million contracts traded per day on average, up 19% from 2020, of which 1.4 million options were S&P500 options and 522,000 were VIX options. VIX futures traded on average 215,000 contracts daily in 2021.

Additional indices exist including the short-term volatility index (VIX9D) and longer-term volatility indices (VIX3M, VIX6M, VIX1Y). There are also volatility indices for other equity indices such as the Nasdaq-100 (VXN), Dow Jones Industrial Average (VXD), Russell 2000 (RVX) and Emerging Markets ETF (VXEEM).

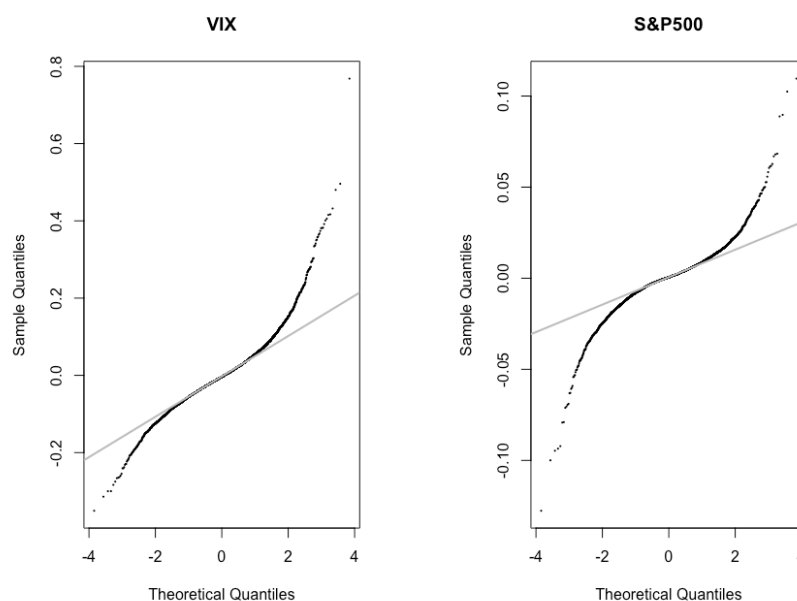


Figure 2: *Q-Q plot of the logarithmic returns of the VIX and S&P500 (1990-2022).*

There are also a number of Exchange Traded Funds (ETF) like the ProShares VIX short-term and mid-term futures ETF (VIXY, VIXM), short VIX short-term SVXY and ultra VIX short-term futures ETF (SVXY, UVXY) and also Exchange traded Notes (ETN) such as the VelocityShares Daily 2x short-term ETN (TVIXF) and iPath Series B S&P500 VIX short-term and mid-term futures ETN (VXX, VXZ) to name a few. These offer investors the convenience of trading options on the VIX without having to manage the options themselves directly with a focus on being long or short VIX.

It can be shown that volatility and the VIX index are negatively correlated to the equity market and just like the equity market the distribution of returns do not fit a normal distribution. In Figure 1 the history of S&P500 versus VIX is plotted. In Figure 2 the Q-Q plots of VIX and S&P500 returns are depicted, the deviation from a normal distribution is apparent. One advantage of the VIX over the S&P500 index though is that the VIX is mean-reverting. We do not have to deal with detrending the time series or returns for that matter. Figure 3 shows the returns of the VIX versus the S&P500. In all cases the logarithmic returns are used, since that normalizes the up- and down moves of the asset.

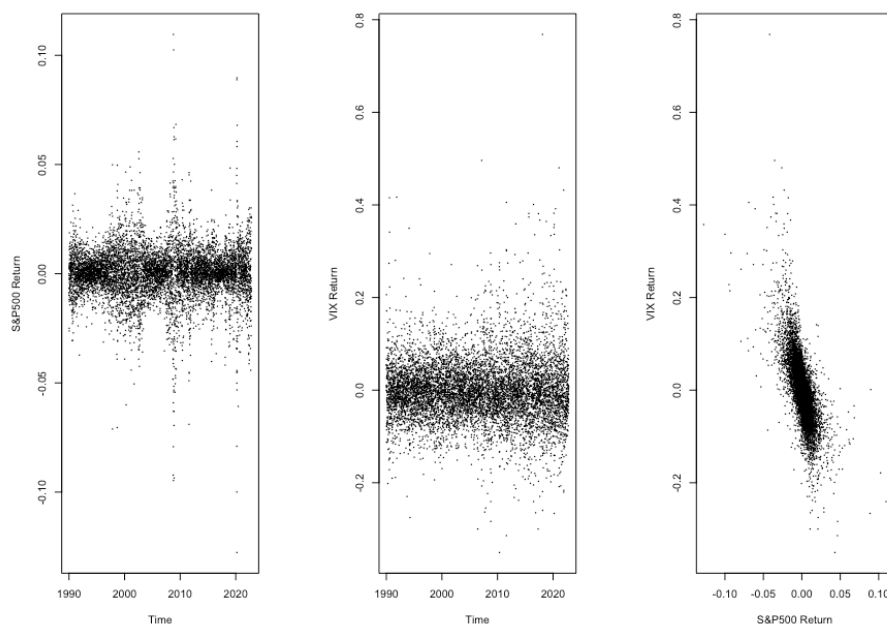


Figure 3: Plot of the S&P500 (left) and VIX (center) logarithmic returns. Plot of the S&P500 returns versus the VIX returns (right).

3 – The Hidden Markov Model and the Gaussian Mixture Model

A Hidden Markov model is defined by a set of transition probabilities A , emission probabilities B and starting probabilities π

$$\lambda = (A, B, \pi) \quad (4)$$

In addition, there is the number of states N .

A Hidden Markov model describes a sequence of observations that are either discrete or continuous, the observations can have one or multiple dimensions. The *transition*

probability matrix A is a $n \times n$ dimensional matrix describing the probabilities of switching from one state to the next. For a first order Markov model the states of the next observation are described entirely by the current states and the transition probability matrix A . Higher order Markov models are dependent on the current and preceding states up to the number of orders. Here we are only considering the most common approach of using an order one Markov Model.

The actual observations are described by the *emission probabilities matrix* B which maps the states to the observations.

For the case of a continuous observation [18, 52, 68-70] the observations are described by a probability density function (pdf) instead of a matrix B for the discrete case. The general form of such a pdf is a finite mixture, consisting of a sum of Gaussians, the most commonly used distribution in practice

$$b(\mathbf{o}_t) = \sum_{k=1}^N \pi_k \cdot \mathcal{N}(\mathbf{o}_t | \mu_k, \sigma_k) \quad (5)$$

where $\mathcal{N}(\mathbf{o}_t | \mu_k, \sigma_k)$ is the normal distribution with mean μ_k and standard deviation σ_k and the weights π_k sum to 1

$$\sum_{k=1}^N \pi_k = 1 \quad (6)$$

In general, one can express each state by a finite mixture of Gaussians, then Equation (5) becomes a representation of each state. Here we will think of each component of the finite mixture corresponding to one state only [2].

As is suggested in [69] this can easily be generalized to two or more dimensions introducing the covariance σ_{xy_k} . Here we only focus on the one-dimensional case, so only end up with a sum over N different Gaussians with weights that sum up to 1.

The Hidden Markov Model, or in this case now the Gaussian Mixture Model, is defined by

$$\lambda = (\boldsymbol{\mu}, \boldsymbol{\sigma}, \boldsymbol{\pi}) \quad (7)$$

instead of the transition and emission probability matrices it becomes a task of finding the best fitting tuple for the means μ_k , standard deviations σ_k and weights π_k [20].

This is done using the Expectation-Minimization (EM) method. There are several works outlining how this works in detail. Here we will outline the general method only. One starts with the probability, or likelihood, of an observation stemming from a particular Gaussian component. In the minimization step (M) one updates all the components based on the previous likelihood computations. At each iteration of this step the likelihood is computed which in turn updates the components for the following step. The iteration terminates when the likelihood converges to a maximum, see [69, 70].

The parameter tuple of means μ_k , standard deviations σ_k and weights π_k can be written in terms of probabilities [71-73]

$$\mu_{jk} = \frac{\sum_{t=1}^T \gamma_t(j, k) \cdot \mathbf{o}_t}{\sum_{t=1}^T \gamma_t(j, k)} \quad (8)$$

for the means

$$\sigma_{jk}^2 = \frac{\sum_{t=1}^T \gamma_t(j, k) \cdot (\mathbf{o}_t - \mu_{jk})(\mathbf{o}_t - \mu_{jk})'}{\sum_{t=1}^T \gamma_t(j, k)} \quad (9)$$

for the standard deviations, and

$$\pi_{jk} = \frac{\sum_{t=1}^T \gamma_t(j, k)}{\sum_{t=1}^T \sum_{j,k=1}^N \gamma_t(j, k)} \quad (10)$$

the weights of the finite mixture. $\gamma_t(j)$ is the probability of being in state j at time t , prime denotes the vector transpose, and $\gamma_t(j)$ is given by

$$\gamma_t(j, k) = \frac{\alpha_t(j)\beta_t(j)}{\sum_{j=1}^N \alpha_t(j)\beta_t(j)} \cdot \frac{\pi_{jk} \cdot \mathcal{N}(\mathbf{o}_t | \mu_{jk}, \sigma_{jk})}{\sum_{k=1}^N \pi_{jk} \cdot \mathcal{N}(\mathbf{o}_t | \mu_{jk}, \sigma_{jk})} \quad (11)$$

Equation (11) is the ratio of the expectation of being in state j using the k^{th} mixture and the expectation of being in state j . The expectation of the mean and standard deviations Equation (8,9) are weighted similarly by each observation and scaled by the sum of being in state j using the k^{th} mixture. The new terms $\alpha_t(j)$ and $\beta_t(j)$ here, are the forward and backward variables, solved for in the *forward-backward* procedure, as they are used in the common Hidden Markov Model. Solving this system of equations is now defined similar to the discrete case of the Hidden Markov Model. Three steps are commonly identified in this process.

The Hidden Markov Model:

Step 1 (Computing the Likelihood / Encoding):

Given a model and an observation sequence, the probability of the observation sequence needs to be determined, this is solved by what is called the *forward-backward* procedure.

Step 2 (Decoding):

Given a model and the observation sequence, what is the underlying state sequence that best explains the observations. This step is solved by the *Viterbi* algorithm.

Step 3 (Training/ Learning):

This step aims to maximize the probability of the observation sequence given the model. This is done using the Expectation-Maximization (EM) algorithm.

The algorithms used in the Hidden Markov Model are derived at length in the literature and will not be repeated here. A good description of the forward-backward procedure can be found here [1, 2, 74, 75], the Viterbi algorithm is explained in these references [74, 76] and the Expectation-Maximization algorithm is derived here [77, 78].

S&P500 and VIX

The S&P500 and VIX time series are shown in Figure 1, ranging from 1990 to 2022. Inspecting the chart, it is apparent that the VIX corresponds quite clearly to large movements of the S&P500 and it has the advantage of being trendless. Instead of having to deal with a main trend underlying the movement in price, such as is the case for the S&P500 or the sine wave in the example above, the VIX itself doesn't exhibit this, which will benefit the mixture model, since the time series won't have to be detrended. As is usual the log returns of the VIX are used in the time series because it normalizes the up and down moves in price, giving both directions an equal weighting.

A QQ plot of the S&P500 and VIX is shown in Figure 2 revealing the non-normal characteristic of the S&P500 and VIX.

This is an already well-known aspect of the S&P500; the advantage of a mixture model is that it allows the modelling of non-normal distributions. A single Gaussian won't be able to fit a non-normal distribution, but a series of Gaussians will be able to achieve this. In order to determine the optimal number of states we calculate the Log Likelihood, AIC and BIC with number of states ranging from 1 to 40. Results are shown in Figure 4, showing that the Log Likelihood asymptotically approaches a slope, indicating that there is an increasing accuracy of the model using 4 states and above.

The AIC and BIC show a similar shape and indicate that there is a local minimum around 4-5 for the BIC and 5-10 for the AIC. As is expected the AIC and BIC indicate that there is some overfitting happening for higher number of states. The BIC even suggests that around 4-5 states is the ideal number to use. Underfitting happens when less than 4 states are used also confirming our choice of 4 states. Judging from the Log Likelihood, AIC, BIC the use of 4 states is a good choice and will be used for the rest of this study.

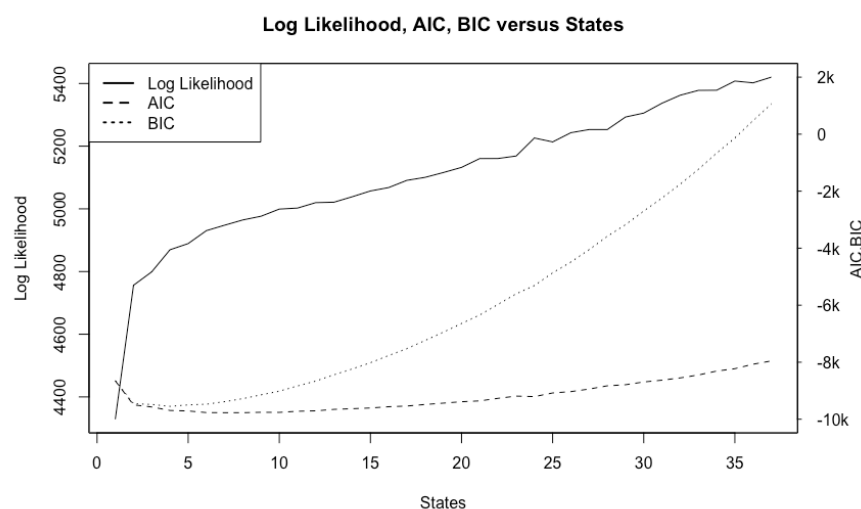


Figure 4: Plot of the Log Likelihood, AIC and BIC values versus increasing number of states of the model.

Figure 5 (top) shows a normal distribution together with the Gaussian Mixture, consisting of the sum of 4 normal distributions, Equation (5) and Figure 5 (bottom) the corresponding Gaussian mixture components. Notice how this mixture has a higher probability density for bigger positive moves and deviates from the normal distribution in various places.

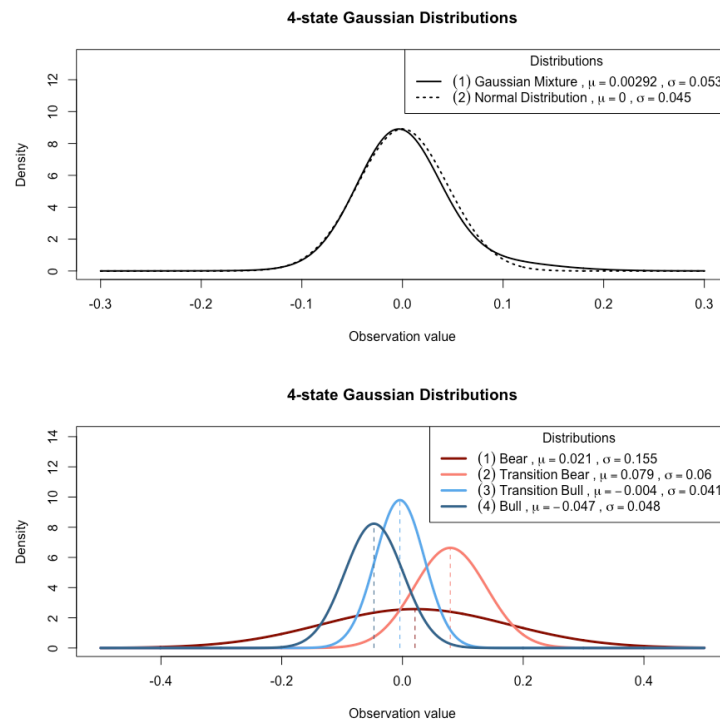


Figure 5: Comparison of a normal distribution and a Gaussian mixture (top). Decomposition of a 4-state Gaussian mixture into the 4 normal distributions (bottom).

The 4-state Gaussian mixture shown here is the actual result of the Hidden Markov Model using 4 Gaussians, where the entire time series from 1990 to 2022 is used to fit the model.

In practice each state has to be uniquely identified, labelling them the same way from one set of data (or time step) to the next. This is done by introducing the following measure, the product of mean and variance

$$\mu \cdot \sigma^2 \quad (12)$$

This measure sorts the states from the most positive means with the biggest standard deviations squared to the most negative means in a descending order. Positive means correspond to increases in the VIX volatility index and negative means to decreases. This measure therefore ranks the states from the most bearish to bullish in a systematic way. An N-state Gaussian mixture would result in a spectrum of states from bearish to bullish. Since we are only using 4-states and for the sake of simplicity we will abbreviate the 4 states as *bear*, *transition bear*, *transition bull* and *bull*.

Inspecting a regime plot, Figure 6, where all the 4 probabilities are stacked on top of each other it is noticeable that the bearish states are usually most pronounced in periods of crisis such as the pandemic selloff in 2020 or the GFC of 2008/09, whereas in periods of calm the bullish states have predominance. But this is not always the case, since inspecting Q4 of 2021 the bearish states have dominance even though the selloff accelerated from Q1 of 2022 only, highlighting that these states describe the VIX movements during this period and is better viewed in terms of the ranking, Equation (14), instead of the bull/bear terminology. Since these states though predominantly correspond to times of distress we will stick with the existing labelling.

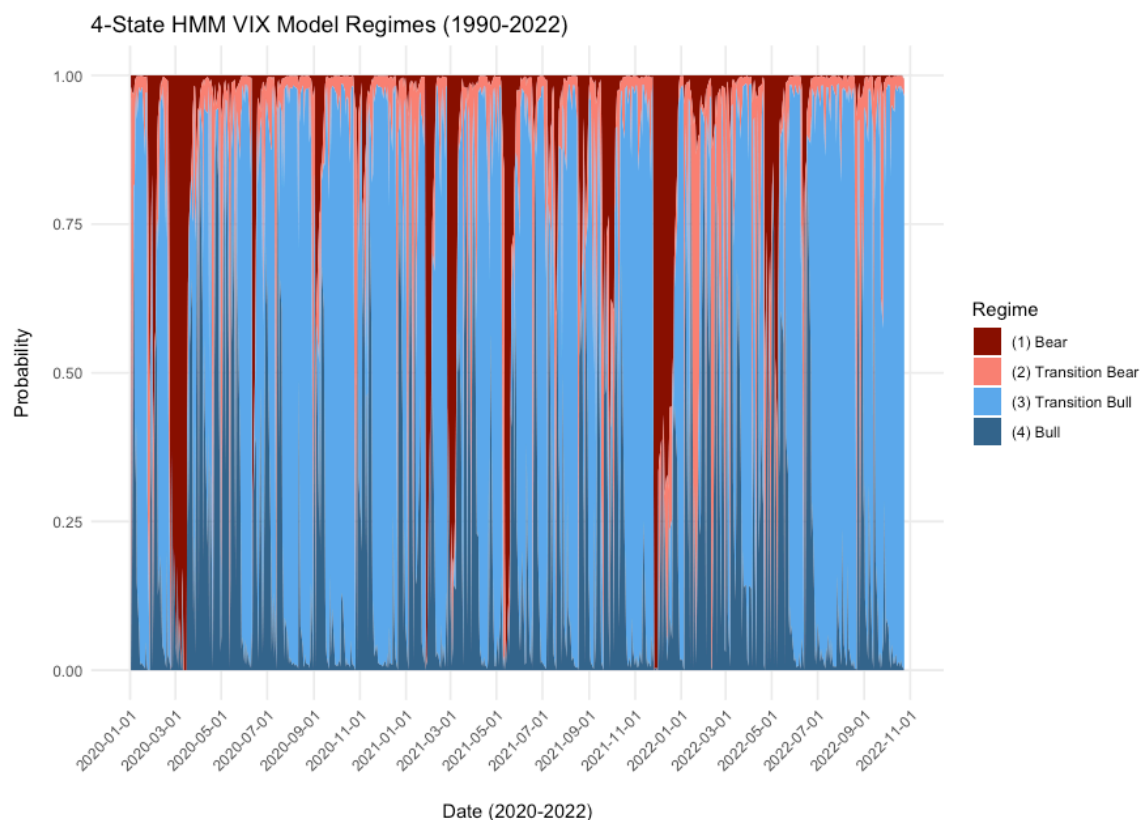


Figure 6: Regime plot of the probabilities of the 4-state Gaussian Mixture Hidden Markov Model for the VIX (showing only 2020-2022).

4 – Trading Strategy

A trading strategy always involves a prediction for a future unknown data point. The strategy decides on taking a long or short position or no position at all. Hidden Markov Models have been used to develop such trading strategies. After having fitted the model additional methods are commonly employed to make the prediction.

[18, 20, 24] use the expected mean of the Gaussian to determine the direction of the position, with and without a threshold level. [47] uses the likelihood to match patterns with the past and extrapolate from past patterns. Fuzzy Logic, Artificial Neural Networks

(ANN), Genetic Algorithms (GA) and further examples of methods employed utilizing the results of the HMM [48, 51, 79, 80].

Here the results of the Gaussian mixture model fitted to the VIX using the Hidden Markov Model are used to determine the trading direction on the S&P500. The VIX and the S&P500 are indirectly linked to each other. The S&P500 represents the spot index of all stocks within it, while the VIX represents the risk-neutral expectation of the 30-day realized volatility of the index. The VIX itself incorporates prices of a wide range of volatility options along many strikes, covering many hundred different options across the market. It has been recognized that the price of the VIX contains a risk premium that is not yet reflected in the price of the S&P500 itself, from which predictions for the S&P500 could be derived from [65]. The Hidden Markov Model is aiming to seek out this additional information hidden in the time-series of the volatility asset.

Fitting a Gaussian Mixture model to a given set of data not only allows us to identify the states in the past, but also yields the predicted probabilities for the step-ahead.

The predicted probabilities $\mathbf{p}$ at every step ahead is given by the product of the posterior probabilities $\boldsymbol{\pi}$ (the coefficients of the Gaussian mixture) and the transition probability matrix $\mathbf{A}$ (in this case a 4x4 matrix)

$$\mathbf{p} = \mathbf{A} \cdot \boldsymbol{\pi} \quad (13)$$

Since the step-ahead probability vector $\mathbf{p}$ is the coefficient vector of the Gaussian Mixture, the expected mean E_{Mean} one step ahead is the vector product of $\mathbf{p}$ and the vector of means of the Gaussian Mixture

$$E_{Mean} = \mathbf{p} \cdot \boldsymbol{\mu} = \sum_{k=1}^N p_k \cdot \mu_k \quad (14)$$

The expected variance E_{Var} is calculated by calculating the modified expected variance and subtracting the expected mean squared [81]

$$E_{Var} = \sum_{k=1}^N p_k \cdot (\mu_k^2 + \sigma_k^2) - E_{Mean}^2 = E_{STD}^2 \quad (15)$$

taking the square root yields the expected standard deviation E_{STD} .

The long-term state is calculated by taking powers of the transition probability matrix and multiplying it by the posterior probabilities, Equation (15) becomes

$$\mathbf{p}_{LT} = \mathbf{A}^n \cdot \boldsymbol{\pi} \quad (16)$$

The long-term expected mean, expected variance and expected standard deviation are calculated similar to Equations (16) and (17).

In practice a trading strategy exists of a data set of past and current values of the VIX, serving as the seed data to fit using the Gaussian mixture model. Here we start with a set

of 2000 days of seed data, which corresponds to roughly 7.5 years from 1990 to 1998, and a trading range from 1998 to 2022 with about 6266 trading days. The Gaussian Mixture Hidden Markov model is calculated for the window of seed data for every new trading date starting from the 2000th. At the end of each window we calculate the predicted probabilities $\mathbf{p}$, long term predicted probabilities $\mathbf{p}_{LT}$, expected mean E_{Mean} and standard deviation E_{STD} as described above. For every new trading date following we extend the existing seed data by the additional known data point and run the same Gaussian mixture model again, repeating the steps as before. This way we receive the set of prediction values for every single time step in the future until we reach the last known date of the VIX. The window of data that is being used to model the data is therefore extending on one end, the beginning of the window is fixed. The calculations are done in R using the depmixS4 package [82, 83], which takes 2.5 days to compute on an Apple M1 (gen1) (the M1 has a single-core CPU speed roughly 82% of the fastest CPU available to date and the calculation doesn't allow multi-core utilization).

At the end of each window one has the expected mean E_{Mean} and expected standard deviation E_{STD} of the following step. Together with the current state of the model and the associated predicted probabilities $\mathbf{p}$, one can plot the expected mean versus the expected standard deviation, as in Figure 7, where the thickness and color of each point corresponds to the state and its associated predicted probability.

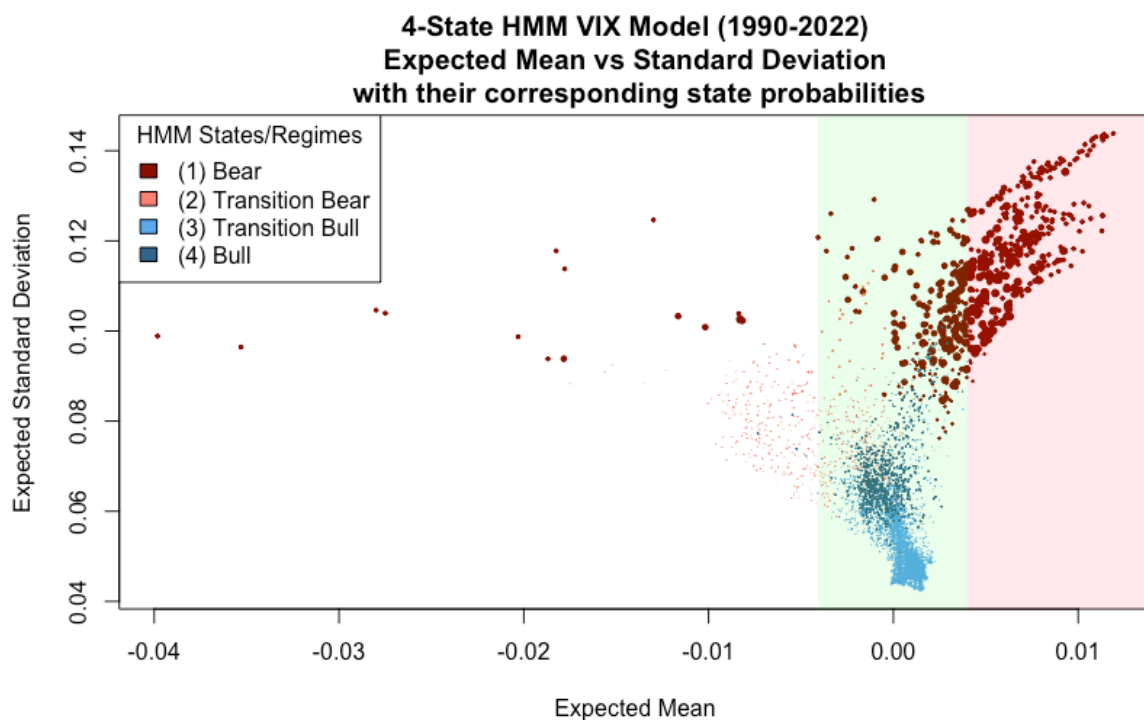


Figure 7: Scatter plot of the predicted probabilities (as weights) and corresponding states of the 4-state Gaussian Mixture HMM (1990-2022). Bear State (1) in dark red, Transition Bear (2) in salmon, Transition Bull (3) in blue and Bull (4) in dark blue. Shaded regions correspond to buy zones (green) and sell zones (red).

The 4 different states are grouped together in 4 different regions of this space. The majority of bear states have increasing expected standard deviation with increasing expected means and probabilities. The transition bull state exhibits decreasing standard deviation with increasing expected means and probabilities. The bull states have slightly

negative means and small expected standard deviations and the transition bear states are distributed across the widest range of negative expected means. From the probabilities in this chart it becomes apparent that the bear case, the transition bull and bull case have the highest probability weights.

One trading strategy (Strategy 1) optimizes the performance using the expected mean. Finding the optimal cutoff for the expected mean in this case yields 0.0041. When the absolute value of the expected mean is smaller than this cutoff one takes a positive position for the following day. This area is highlighted in green in Figure 7. When the expected mean is bigger than this cutoff one takes a short position for the following day. This area is highlighted in red in Figure 7. In this approach one focuses solely on the expected mean. Looking at the grouping of the different states it is possible that an optimization in the expected standard deviation improves the performance.

An alternative trading strategy (Strategy 2) is to only take a short position when the bear state exceeds a certain probability. The optimal value for the bear state probability is 0.688. When the value is breached one takes a short position for the following day. In all other cases one takes a long position.



Figure 8: Performance study of the two trading strategies. Strategy 1 on the left and strategy 2 on the right, compared to a Buy-n-Hold of S&P500 (1990-2022)

Results for the two trading strategies are shown in Figure 8 and Table 1, where some of the common performance measures are calculated. Both strategies outperform a regular buy and hold strategy by about 35% on average. Drawdowns are similar to a Buy-n-Hold although drawdown lengths are larger.

Table 1: Trading statistics of the two strategies (1990-2022).

	Buy-n-Hold	Strategy 1	Strategy 2
Annualized Return	3.67%	5.21%	4.68%
Average Drawdown	2.36%	2.77%	2.67%
Average Length	36.64	49.28	45.15
Average Recovery	15.63	23.63	21.39
Annualized Std Dev	18.3%	19.41%	19.80%

In another study we will use the same methodology to look at a data set of about half the size, 4266 data points starting from 2005 onwards. Similar to before the seed data consists of the first 2000 dates and the window of seed data grows at every new trading date added until we reach the current date of 2022.

The expected mean E_{Mean} and expected standard deviation E_{STD} for this data set is plotted in Figure 9. The bear case doesn't span as wide of a range of expected means being more concentrated and the bull and transition bull cases are distributed similar to the full range data set 1990-2022. The bull case is distributed around slightly negative means and the transition bull case stretching to positive means with decreasing standard deviations.

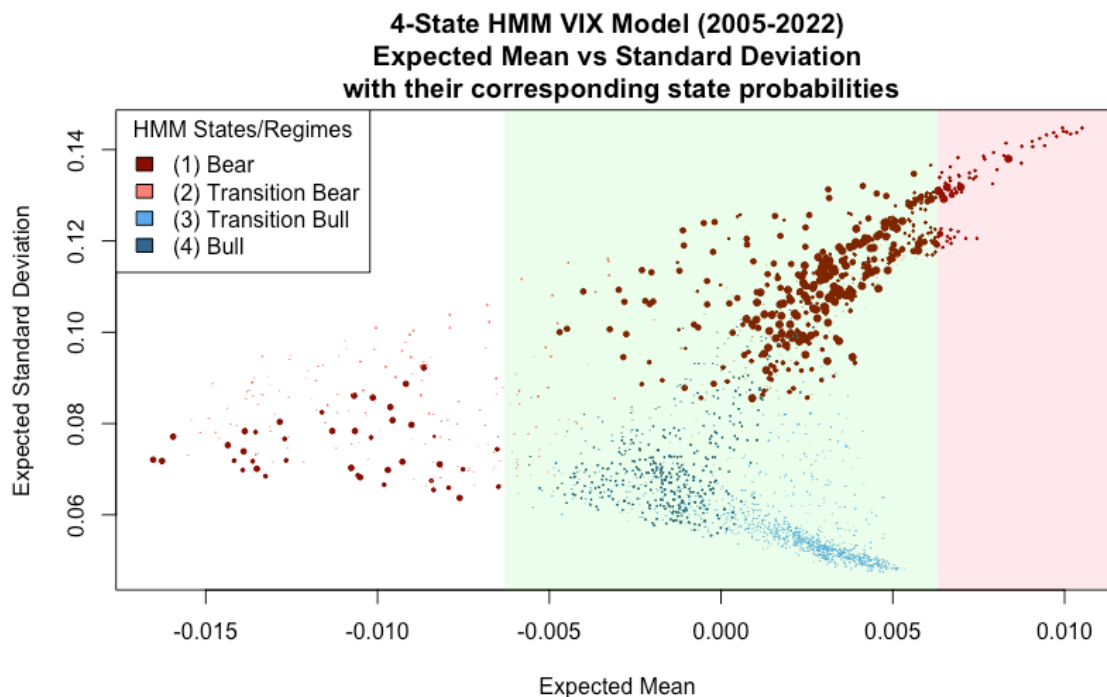


Figure 9: Scatter plot of the predicted probabilities (as weights) and corresponding states of the 4-state Gaussian Mixture HMM (2005-2022). Bear State (1) in dark red, Transition Bear (2) in salmon, Transition Bull (3) in blue and Bull (4) in dark blue. Shaded regions correspond to buy zones (green) and sell zones (red).

When we apply the trading strategies (1) and (2) we find an optimal cutoff for the expected mean of 0.0063 for strategy (1) and 0.61 for the cutoff of the bear state probability of strategy (2). The performance of the two strategies are plotted in Figure 10 comparing it to a Buy-n-Hold of the S&P as well as the Cboe VIX Tail Hedge Index (VXTH).

The VXTH represents a portfolio of the total return of the S&P500 and a dynamic position of a 30-delta call option on the VIX. This 30-delta call represents a tail hedge on the portfolio and is only bought when the one-month forward value of the VIX is between 15% and 50%. The amount bought varies between 0.5% and 1% of the portfolio value for a VIX between 30%-50% and 15%-30% respectively. This tail hedge is put on and rolled on a monthly basis.



Figure 10: Performance study of the two trading strategies. Strategy 1 on the left and strategy 2 on the right, compared to a Buy-n-Hold of S&P500 and VXTH (2005-2022)

The performance statistics of both strategies and the VXTH shown in Figure 10 can be found in Table 2. What is noticeable is that the best performance for both strategies as well as the VXTH is around the biggest drops in the index or the highest levels of the VIX, during the pandemic selloff in March 2020. This is also already apparent when looking at the full data set of Figure 8. The VXTH is expected to have an outsized gain during such periods considering that the call option on the VIX pays off handsomely. It is interesting to note that the two trading strategies indicate similar concentration of gains during the same period.

In other periods of calmer markets all three strategies have varying degrees of success, with strategy (2) performing better and even outperforming the VXTH strategy too.

Strategy (1) and (2) outperform a Buy-n-Hold strategy by 59% on average. Comparing this to the VXTH, considering that the VXTH portfolio consists of the total return (with dividends reinvested), suggests that both strategies beat the VXTH by about 2-3% in annualized return. Drawdowns and lengths are very similar to each other. The annualized standard deviation of both strategies is similar to that of the Buy-n-Hold.

Table 2: Trading statistics of the two strategies (2005-2022).

	Buy-n-Hold	VXTH	Strategy 1	Strategy 2
Annualized Return	7.27%	10.93%	12.06%	11.20%
Average Drawdown	1.79 %	1.91%	2.05%	2.38 %
Average Length	14.07	19.18	16.68	18.68
Average Recovery	6.94	9.18	7.84	8.15
Annualized Std Dev	18.0%	15.03%	17.53%	17.98%

The sensitivity of the cutoff parameters in both studies are reasonably well behaved, with R^2 values of 63% and 77% using a parabolic fit. The full data set being noisier than the data set from 2005 to 2022.

Comparing the full data set from 1990 to 2022 to the data set of half the size from 2005 to 2022 suggests that in both cases the trading strategy outperforms a Buy-n-Hold. However, the data set with more recent data is doing better than the full data set. The period 2010-2020 in particular is much worse when compared to the second data set. This is most likely due to a number of factors affecting the underlying data outlined in the following.

Firstly, the composition of the S&P500 changes with constituents growing or being added to or dropped from the index. While Telecom, Utilities, Materials and Energy made up more than 35% of the index in 1990, they represent less than 20% today and the index is being dominated by a few companies only in 2022. One of the requirements of the Hidden Markov Model is that the hidden process itself is not varying. In the example of the set of coins, it would be akin to repeatedly modifying or swapping out the coins, changing the nature of the whole experiment. In fact, most other volatility modelling approaches will have difficulties accommodating such recurring modifications.

Secondly, while the data for the VIX may be available going back to 1990, it was only based on the S&P500 in 2003, futures and options were introduced in 2004 and 2006 respectively, with weekly options were included in its calculation only in 2014. The establishment of a fully developed market on the VIX can be recognized with the onset of these instruments as well as the large increase in volume traded.

For the reasons mentioned above the successful application of the Gaussian Mixture Hidden Markov Model will suffer due to the inconsistent nature of the input data.

Considering these short comings of the data as a source it actually performs reasonably well even on the full data set 1990-2022. It would be interesting to apply this model to single stock data, calculating a single-name VIX equivalent and using the single name as the underlier in the trading strategy.

It is also worth noting that the average daily volume on the VXTH is very low and replicating the same strategy manually will pose the problem of deciding on when to put a tail hedge in place. A monthly roll of the tail hedge leaves a lot of time to be potentially un-/protected. An algorithm that predicts the optimal bear case scenarios would assist in such a tail hedge strategy and maybe the HMM can do just that.

The long term predicted probabilities p_{LT} calculated at the end of each window of data is shown in Figure 11. What is noticeable is that the transition bull and bull state probabilities make up around 75% of all probabilities, with the transition bear and bear probabilities accounting for 25%. Within those the bull state probability makes up around 20% and the bear state probability around half that.

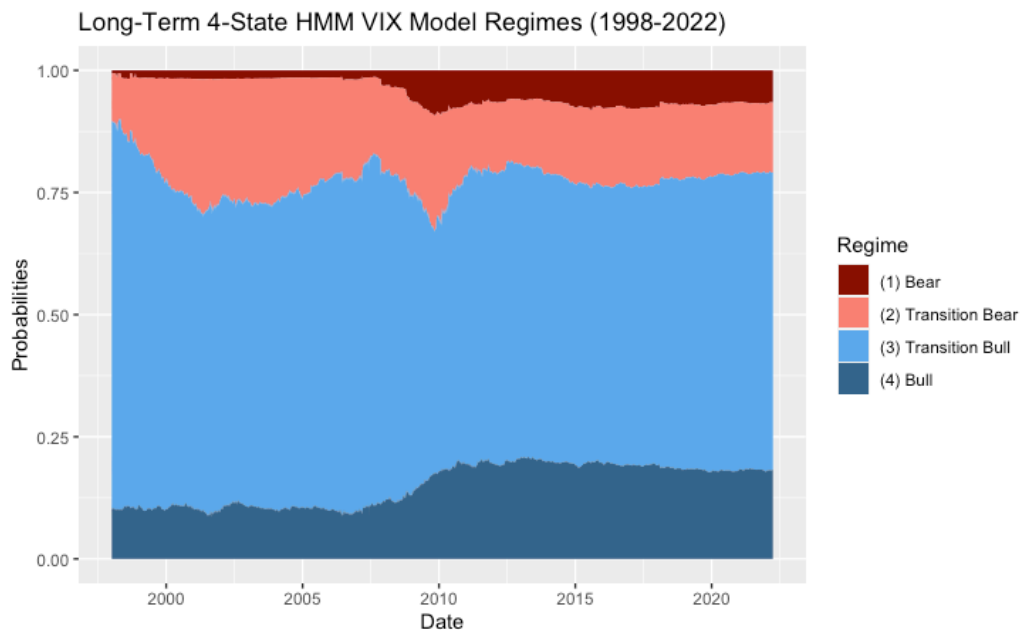


Figure 11: Long-term predicted probabilities of the 4 states (1998-2022).

In summary a trading strategy using the Gaussian mixture model involves the following steps

Trading Strategy

1. Specify sufficient seed data for the Gaussian Mixture model
2. Fit the Gaussian mixture model using the Hidden Markov model method to the current window of data, with a fixed start date
3. Identify the same states as in previous step using the measure defined by Equation 14
4. Calculate the predicted probabilities $\mathbf{p}$, expected mean E_{Mean} and standard deviation E_{STD}
5. Determine the optimal cutoff probabilities and/or cutoff expected mean
6. Take a position on the underlying asset for the following day based on Step 5.

Steps 1-6 are repeated for every new data point keeping the beginning of the seed data fixed and extending it by every new data point known. Keeping the beginning of the seed data fixed is important because it will ensure that the Hidden Markov model will always have the same data available as in the previous step and will not lose any information it has learned from previous steps. The model will only get more accurate the more data points are known.

As an additional step in fitting the Gaussian mixture model in step 2, one can repeat the fitting multiple times and only take the model that fits the data with the highest log Likelihood. This is not absolutely necessary but is done here to ensure that one always achieves the best fitting model. Step 5 can be calculated by simple brute force at very little cost as is done here, or by other linear optimization methods. Alternatively, one can apply additional machine learning techniques to find other more optimal trading strategies as mentioned above in the references therein.

5 – Conclusion

A continuous Hidden Markov Model for the VIX is introduced using Gaussian mixtures, presenting a new methodology to modelling and predicting volatility assets. The construction of the VIX is described and its importance to the S&P500 is explained. The results of the Hidden Markov Model for the VIX are used to propose two successful trading strategies on the S&P500, using the predicted state probabilities, expected mean and standard deviation of the time series.

This methodology can be generalized to any volatility surface and seems particularly useful to trading desks which maintain individual volatility surfaces for many assets besides the S&P500 but usually don't store any time series of them. Consequently, there is a lot of data potentially available which remains underutilized today. The competitive advantage of predicting a drop or rise in volatility is extremely valuable to not only volatility traders, but to trading desks as a whole, since it allows for better positioning and improved gains.

Furthermore, the Gaussian Mixture Hidden Markov Model offers an alternative method for modelling volatility and it would be interesting to compare it to other common volatility models (ARCH-type, etc.) measuring its in-sample fit and out-of-sample forecast performance.

There are also various other directions for further research. One could look at other single volatility indices, or apply this model to a multi-dimensional HMM using more than one volatility index introducing covariance. Looking at implied volatility surface data from the options market is another obvious place to look for success. Additionally, there are other methods one could employ to improve on the trading performance such as neural networks. Optimizing the cutoff expected means, standard deviations and predicted probabilities is another avenue for improvement.

The author would like to thank the reviewers for their helpful comments and suggestions.

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